

TAWOS DATASET

Can planning-time Jira data reveal which sprints will struggle?

TAWOS is a public dataset of Agile open-source software projects collected from Jira repositories. We selected eight projects because the dataset directly supports our research on predicting late sprint closure and unfinished committed scope from information available at Sprint Planning.

The key challenge: reconstructing what the team actually knew at sprint start without using future information.

2,657

original sprint records

145,309

issues

8

open-source projects

Two prediction tasks

1

Late sprint closure

Sprint closes >1 day after its planned end.

2

Unfinished committed scope

At least one sprint-start issue remains unresolved at the planned end.

Core tables

Sprint

Issue

Change_Log

Maryam Hanif • Hani AbuSharkh • Samra Nawazish • Najwa Suleiman • Alyah Alameeri

GitHub: github.com/hani-ziad/SwEng_202610

Handling the Data

Reconstruct the planning-time state first; model only what was knowable then



1 FILTER

Remove FUTURE and ACTIVE sprints; remove impossible timestamps; exclude one genuine cold start per project.

2 RECONSTRUCT

Walk Jira change history backward to sprint start. Rebuild Sprint membership and Story Points at that timestamp.

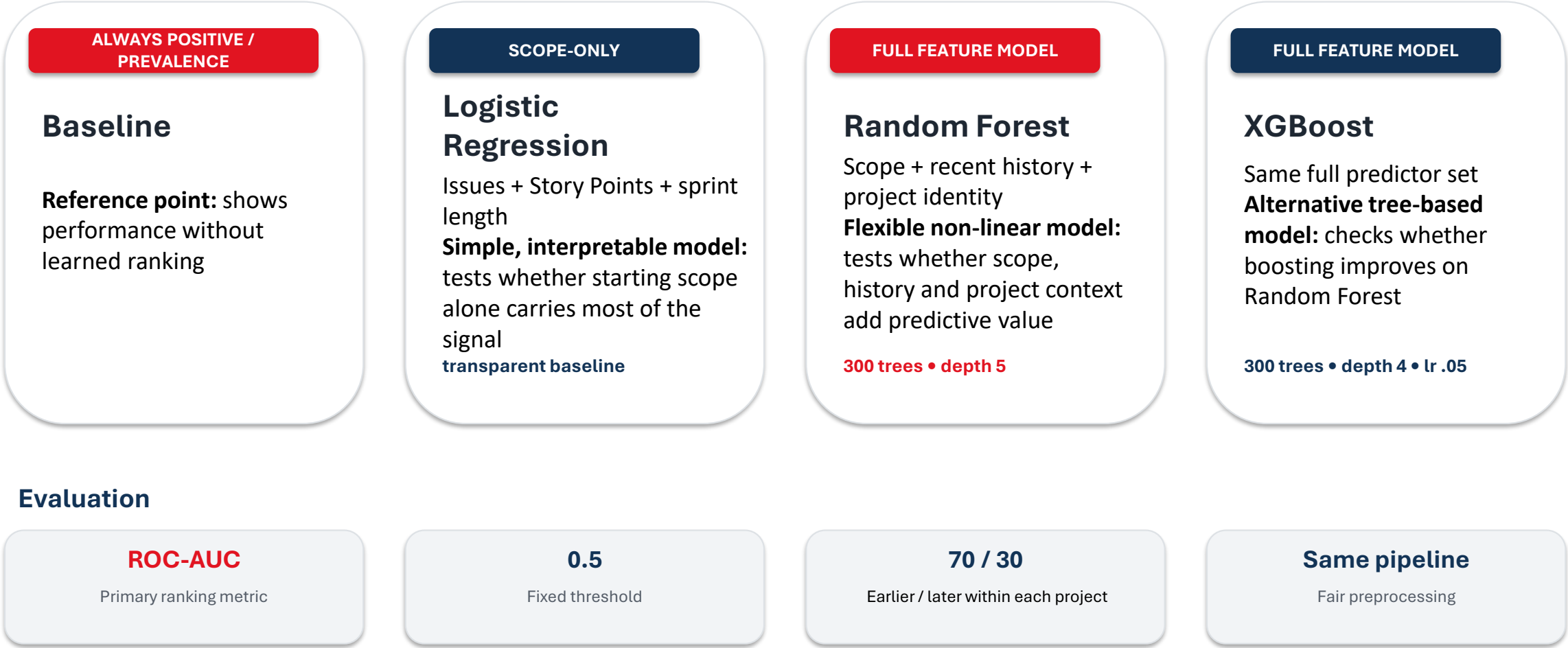
3 PREVENT LEAKAGE

Use only information available at sprint start.
Main evaluation: chronological split within each project - earlier 70% train, later 30% test.

Main data problem

Some issues had missing Sprint or Story Points change history. We used the static Jira value as a documented fallback and later tested how this assumption affected the results.

Simple baselines and flexible tree models are evaluated under the same time-ordered framework



Also reported: F1 • Precision • Recall • PR-AUC • Brier score

Results

The two outcomes behave differently: late closure needs context; unfinished scope is mostly a scope story

Outcome	Model	F1	ROC-AUC	Train time	Model size / setup
Late closure	Always positive	.477	.500	0 s	No fitted model
Late closure	Scope-only LR	.336	.530	0.0202 s	3 predictors
Late closure	Random Forest	.562	.705	1.2510 s	300 trees
Late closure	XGBoost	.535	.695	1.7950 s	300 trees
Unfinished scope	Always positive	.971	.500	0 s	No fitted model
Unfinished scope	Count-only LR	.650	.881	0.0198 s	1 predictor
Unfinished scope	Scope-only LR	.696	.885	0.0195 s	3 predictors
Unfinished scope	Random Forest	.875	.852	1.1374 s	300 trees

LATE CLOSURE

0.705 AUC

Random Forest gives the strongest ranking performance; XGBoost is close at 0.695.

UNFINISHED SCOPE

0.885 AUC

Scope-only Logistic Regression gives the strongest ranking performance, above RF at 0.852.

Note: Unfinished-scope results use nonempty sprints (n = 511). Training times are measured model-fit times from the reproducible run on this machine.

Why? Explanation + Ablation

The strongest result is explained by a property of the data, not by model complexity

Late closure: Random Forest performed best (AUC.705)

Signal is weaker and more project-specific.

7.3 → 8.8

mean issues

32.9 → 38.3

mean story points

Project identity +
sprint length
contribute most.

Unfinished scope: Scope-only Logistic Regression performed best (AUC .885)

Initial workload dominates.

-0.220 AUC

when committed issue count is shuffled

Next-largest drop: only
0.021

Feature-group ablation

Random Forest features	Late closure AUC	Full spillover AUC
Full	.705	.986
Planning only	.706	.987
History only	.625	.697
No project identity	.696	.989

What the ablation shows

Removing planning-time scope collapses spillover AUC from .986 to .697. Keeping planning features alone leaves it essentially unchanged (.987).

Conclusion: starting scope carries most of the measurable unfinished-work signal; late closure remains weaker and more context-dependent.